Graphing Quadratics from Vertex or Standard Form Answer Key

Algebra 1 Parabola Features and Transformations

Quick Answers

1. -1
2. 4
3. -4, 5
4. -1, 5
5. -8, -1, 3
6. 9, 5
7. -3, 3
8. -2, -4, 0
9. -4, -3, 1
10. 1, 3, -2
11. $(x-3)^2-5$, -5
12. 0

Curriculum

Level: Algebra 1 Parabola Features and Transformations
HSA-SSE.B.3 – *problems 5, 11*
HSF-IF.A–HSF-IF.C – *problems 1, 2, 3, 4, 6, 7, 8, 9, 10, 12*
Difficulty 1–5 of 5 across 20 tagged checks.

Common wrong answers

1. *If they got 15: read the sign of h straight off the parentheses, giving a vertex x of -2 and a y-value of 15*
3. *If they got 32: used x = b/2a instead of -b/2a, landing on x = -3*
6. *If they got -7: dropped the minus sign on a when computing -b/2a, landing on x = -2*

1. $y=(x-2)^2-1$.

Step 1. This is vertex form $y=a(x-h)^2+k$ with $a=1$, $h=2$, $k=-1$. The subtraction inside the parentheses is what makes h positive.
Step 2. Check by substituting the axis value: $y(2)=(2-2)^2-1=-1$, which is the lowest point because $a=1>0$.
Step 3. Vertex $\boxed{(2,\,-1)}$, opening upward.
Sketch. Axis $x=2$; y-intercept $y(0)=3$; by symmetry the curve also passes through $(4,3)$, and the x-intercepts are 1 and 3.
Common wrong answer: a vertex y-value of 15 — the sign of h was read straight off the parentheses, putting the vertex at $x=-2$.

2. $y = -(x+3)^2 + 4$.
Step 1. Rewrite the inside as $(x - (-3))$, so $h = -3$ and $k = 4$. The leading coefficient is -1.
Step 2. $y(-3) = -(0)^2 + 4 = 4$, and because $a = -1 < 0$ this point is a maximum: the parabola opens downward.
Step 3. Vertex $\boxed{(-3,\ 4)}$, opening downward.
Sketch. Axis $x = -3$; y-intercept $y(0) = -9 + 4 = -5$; symmetric point $(-6, -5)$; x-intercepts at -5 and -1.

3. $y = x^2 - 6x + 5$.
Step 1. Standard form with $a = 1$, $b = -6$, $c = 5$. The axis of symmetry is $x = -\dfrac{b}{2a} = -\dfrac{-6}{2} = 3$.
Step 2. Substitute to get the vertex y-value: $y(3) = 9 - 18 + 5 = -4$.
Step 3. Vertex $\boxed{(3,\ -4)}$; the y-intercept is c, that is $y(0) = \boxed{5}$
Sketch. Opens up; x-intercepts from $(x-1)(x-5)$ are 1 and 5, and they straddle $x = 3$ as they must.
Common wrong answer: 32 — $x = \frac{b}{2a}$ was used instead of $-\frac{b}{2a}$, putting the axis at $x = -3$.

4. $y = x^2 - 4x - 5$.
Step 1. The x-intercepts are where $y = 0$, so factor: two numbers multiplying to -5 and adding to -4 are -5 and 1.
Step 2. $x^2 - 4x - 5 = (x-5)(x+1)$, so $x - 5 = 0$ or $x + 1 = 0$.
Step 3. x-intercepts $\boxed{x = -1 \text{ and } x = 5}$
Sketch. The axis of symmetry is midway between them, at $x = 2$, and $y(2) = -9$, so the vertex is $(2, -9)$. The y-intercept is -5.

5. $y = 2(x-1)^2 - 8$.
Step 1. Vertex form with $h = 1$, $k = -8$, $a = 2$, so $y(1) = \boxed{-8}$ and the parabola opens up, narrower than $y = x^2$.
Step 2. For the x-intercepts set $y = 0$: $2(x-1)^2 = 8$, so $(x-1)^2 = 4$ and $x - 1 = \pm 2$.
Step 3. x-intercepts $\boxed{x = -1 \text{ and } x = 3}$
Sketch. Vertex $(1, -8)$, axis $x = 1$, y-intercept $y(0) = -6$.
Note: vertex form hands you the intercepts in two moves — no factoring and no quadratic formula.

6. $y = -x^2 + 4x + 5$.
Step 1. Here $a = -1$, $b = 4$, $c = 5$. Axis of symmetry: $x = -\dfrac{4}{2(-1)} = 2$. Keeping the sign of a is the whole difficulty.
Step 2. $y(2) = -4 + 8 + 5 = 9$. Because $a < 0$, this is a *maximum*.
Step 3. Vertex $\boxed{(2,\ 9)}$, a maximum; y-intercept $y(0) = \boxed{5}$
Sketch. Opens down; $-x^2 + 4x + 5 = -(x-5)(x+1)$, so the x-intercepts are -1 and 5.
Common wrong answer: -7 — the minus sign on a was dropped in $-\frac{b}{2a}$, putting the axis at $x = -2$.

7. $y = x^2 - 9$.
Step 1. There is no x term, so $b = 0$ and the axis of symmetry is $x = -\frac{0}{2} = 0$ — the y-axis itself.
Step 2. Set $y = 0$: $x^2 = 9$, a difference of squares $(x-3)(x+3) = 0$.
Step 3. x-intercepts $\boxed{x = -3 \text{ and } x = 3}$
Sketch. Vertex $(0, -9)$ on the y-axis, opening up. The two intercepts are mirror images precisely because the axis is $x = 0$.

8. $y = \frac{1}{2}(x+2)^2 - 2$.
Step 1. Vertex form with $h = -2$, $k = -2$, $a = \frac{1}{2}$. Since $|a| < 1$, the parabola is *wider* than $y = x^2$: at one unit from the axis it rises only $\frac{1}{2}$ instead of 1.
Step 2. $y(-2) = \boxed{-2}$, the minimum. For the intercepts set $y = 0$: $\frac{1}{2}(x+2)^2 = 2$, so $(x+2)^2 = 4$ and $x + 2 = \pm 2$.
Step 3. x-intercepts $\boxed{x = -4 \text{ and } x = 0}$
Sketch. Vertex $(-2, -2)$, axis $x = -2$; the curve passes through the origin, which is one of the two intercepts.

9. $y = x^2 + 2x - 3$.
Step 1. $a = 1$, $b = 2$: the axis is $x = -\dfrac{2}{2} = -1$, and $y(-1) = 1 - 2 - 3 = -4$.
Step 2. Factor for the intercepts: two numbers with product -3 and sum 2 are 3 and -1, giving $(x+3)(x-1)$.
Step 3. Vertex $\boxed{(-1,\ -4)}$; x-intercepts $\boxed{x = -3 \text{ and } x = 1}$
Sketch. Opens up, y-intercept -3. Notice the vertex x-value -1 is the average of -3 and 1 — the two routes agree, which is a free check.

10. $y = 2x^2 - 8x + 6$.
Step 1. Factor the leading coefficient out first: $2(x^2 - 4x + 3) = 2(x-1)(x-3)$. A common factor left in place is where sign errors breed.
Step 2. Setting each factor to zero gives the x-intercepts $\boxed{x = 1 \text{ and } x = 3}$
Step 3. The axis of symmetry is midway between the intercepts, at $x = 2$, and $y(2) = 8 - 16 + 6 = -2$, so the vertex is at $\boxed{(2,\ -2)}$
Sketch. Opens up and is narrow ($a = 2$); the y-intercept is 6.

11. $y = x^2 - 6x + 4$ **— complete the square.**
Step 1. Take half of $b = -6$, giving -3, and square it: 9. Add and subtract that inside the expression: $x^2 - 6x + 9 - 9 + 4$.
Step 2. The first three terms are a perfect square: $(x-3)^2 - 5$. Expanding back gives $x^2 - 6x + 9 - 5 = x^2 - 6x + 4$, so the two forms are the same function.
Step 3. Vertex form: $\boxed{y = (x-3)^2 - 5}$, so the vertex is $(3, \boxed{-5})$
Sketch. Axis $x = 3$, opens up, y-intercept 4; the x-intercepts are $3 \pm \sqrt{5}$, about 0.8 and 5.2.

12. Vertex $(3, -4)$**, through** $(5, 0)$**.** *The sketch and the explanation are open work, flagged for manual review; the model answer below is what full credit contains.*
Step 1. A vertex of $(3, -4)$ fixes h and k, so the equation must look like $y = a(x-3)^2 - 4$. Only a is still unknown, and one more point is exactly enough to find it.
Step 2. Substitute $(5, 0)$: $0 = a(5-3)^2 - 4 = 4a - 4$, so $a = 1$ and the equation is $y = (x-3)^2 - 4$.
Step 3. Check the given point in the finished equation: $(5-3)^2 - 4 = 4 - 4$, so $y = \boxed{0}$ at $x = 5$, as required.
Step 4 (the sketch). Vertex $(3, -4)$, axis $x = 3$, opens up. x-intercepts: $(x-3)^2 = 4$ gives $x = 1$ and $x = 5$. y-intercept $y(0) = 5$.
Full credit: the equation $y = (x-3)^2 - 4$, a sketch labelled with the vertex, the axis $x = 3$ and the intercepts 1 and 5, and a sentence saying that the vertex fixes h and k while the extra point is the only thing that can determine the stretch factor a.